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1.  #include <stdio.h>
2.  #include <string.h>
3.
4.  int main()
5.  {
6.      int marks[5] = {80, 75, 90, 85, 95};
7.
8.      printf("Student Marks\n");
9.
10.     for(int i=0; i<=5; i++)
        // ERROR: Array index goes out of bounds.
        // CORRECTION: for(int i=0; i<5; i++)
11.     {
12.         printf("%d ", marks[i]);
13.     }
14.
15.     printf("\n");
16.
17.     int total = 0;
18.
19.     for(int i=0; i<5; i++)
20.     {
21.         total += marks[i];
22.     }
23.
24.     float average;
25.
26.     average = total / 5;
        // ERROR: Integer division causes loss of precision.
        // CORRECTION: average = (float)total / 5;
27.
28.     printf("Average = %.2f\n", average);
29.
30.     printf("\n");
31.
32.     int numbers[10];
33.
34.     for(int i=1; i<=10; i++)
        // ERROR: Valid indices are 0 to 9.
        // CORRECTION: for(int i=0; i<10; i++)
35.     {
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36.     numbers[i] = i * 10;
37. }
38.
39. printf("Numbers Array\n");
40.
41. for(int i=0; i<10; i++)
42. {
43.     printf("%d ", numbers[i]);
44. }
45.
46. printf("\n");
47.
48. char name[10];
49.
50. strcpy(name, "ComputerScience");
    // ERROR: Destination array is too small.
    // CORRECTION: char name[20];
51.
52. printf("%s\n", name);
53.
54. printf("\n");
55.
56. char city[20] = "Jaipur";
57.
58. city = "Delhi";
    // ERROR: Cannot assign a string to an array.
    // CORRECTION: strcpy(city, "Delhi");
59.
60. printf("%s\n", city);
61.
62. printf("\n");
63.
64. char str1[20] = "Hello";
65. char str2[20] = "Hello";
66.
67. if(str1 == str2)
    // ERROR: Compares addresses, not string contents.
    // CORRECTION: if(strcmp(str1, str2) == 0)
68. {
69.     printf("Strings are Equal\n");
70. }
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71.     else
72.     {
73.         printf("Strings are Different\n");
74.     }
75.
76.     printf("\n");
77.
78.     char password[8];
79.
80.     printf("Enter Password: ");
81.     gets(password);
        // ERROR: gets() is unsafe and removed from C standard.
        // CORRECTION: fgets(password, sizeof(password), stdin);
82.
83.     printf("Password = %s\n", password);
84.
85.     printf("\n");
86.
87.     char firstName[20] = "Robil";
88.     char lastName[20] = "Varshney";
89.
90.     strcat(firstName, lastName);
        // WARNING: Ensure destination array has enough space.
91.
92.     printf("%s\n", firstName);
93.
94.     printf("\n");
95.
96.     char grade[5];
97.
98.     grade[0] = 'A';
99.     grade[1] = '+';
100.
101.    printf("%s\n", grade);
        // ERROR: String is not null-terminated.
        // CORRECTION: grade[2] = '\0';
102.
103.    printf("\n");
104.
105.    int arr[5] = {1,2,3,4,5};
106.
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107. printf("%d\n", arr[-1]);
    // ERROR: Negative array index is invalid.
    // CORRECTION: Use indices 0 to 4 only.
108.
109. printf("\n");
110.
111. char department[10];
112.
113. scanf("%s", department);
    // WARNING: Possible buffer overflow.
    // CORRECTION: scanf("%9s", department);
114.
115. printf("%s\n", department);
116.
117. printf("\n");
118.
119. char email[15] = "student@gmail.com";
    // ERROR: Array size too small for the string.
    // CORRECTION: char email[25] = "student@gmail.com";
120.
121. printf("%s\n", email);
122.
123. printf("\n");
124.
125. int scores[3];
126.
127. scores[0] = 90;
128. scores[1] = 80;
129. scores[2] = 70;
130. scores[3] = 60;
    // ERROR: Index 3 is out of bounds.
    // CORRECTION: Declare int scores[4];
131.
132. printf("%d\n", scores[3]);
    // ERROR: Accesses invalid index.
133.
134. printf("\n");
135.
136. char subject[10] = "Programming";
    // ERROR: Array size too small.
    // CORRECTION: char subject[20] = "Programming";
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137.
138. printf("%s\n", subject);
139.
140. printf("\n");
141.
142. char username[20];
143.
144. printf("Enter Username: ");
145. scanf("%s", username);
146.
147. if(username == "admin")
    // ERROR: String comparison using == is invalid.
    // CORRECTION: if(strcmp(username, "admin") == 0)
148. {
149.     printf("Valid User\n");
150. }
151. else
152. {
153.     printf("Invalid User\n");
154. }
155.
156. printf("\n");
157.
158. char mobile[10];
159.
160. strcpy(mobile, "987654321012");
    // ERROR: Destination array is too small.
    // CORRECTION: char mobile[13];
161.
162. printf("%s\n", mobile);
163.
164. printf("\n");
165.
166. int attendance[5] = {1,1,1,1,1};
167.
168. int present = 0;
169.
170. for(int i=0; i<=5; i++)
    // ERROR: Array index goes out of bounds.
    // CORRECTION: for(int i=0; i<5; i++)
171. {
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172.     present += attendance[i];
173. }
174.
175. printf("Present Days = %d\n", present);
176.
177. printf("\n");
178.
179. char course[20] = "C Programming";
180.
181. printf("%c\n", course[20]);
    // ERROR: Index 20 is outside valid range.
    // CORRECTION: Use indices 0 to 19.
182.
183. printf("\n");
184.
185. char college[20];
186.
187. college[0] = 'A';
188. college[1] = 'C';
189. college[2] = 'E';
190. college[3] = 'I';
191. college[4] = 'T';
192.
193. printf("%s\n", college);
    // ERROR: String is not null-terminated.
    // CORRECTION: college[5] = '\0';
194.
195. printf("\n");
196.
197. int rollNo[5];
198.
199. for(int i=0; i<5; i++)
200. {
201.     scanf("%d", &rollNo[i]);
202. }
203.
204. printf("\nRoll Numbers\n");
205.
206. for(int i=0; i<5; i++)
207. {
208.     printf("%d ", rollNo[i]);
```

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209. }
210.
211. printf("\n");
212.
213. char section[3] = "IT-A";
    // ERROR: Array size too small.
    // CORRECTION: char section[5] = "IT-A";
214.
215. printf("%s\n", section);
216.
217. printf("\n");
218.
219. printf("Program Finished\n");
220.
221. return 0;
222. }
```